

TEAM BRAIN STACK

ET AI-Native News Experience

A Multi-Agent System for Intelligent News Delivery

ECONOMIC TIMES · GEN AI HACKATHON 2026

Submitted by	Team Brain Stack
Event	ET Gen AI Hackathon 2026
Track	Problem Statement #8 — AI-Native News Experience
Solution	Multi-Agent AI News System (5 Agents)
Technology	Python · FastAPI · Groq API · LLaMA 3.3 70B

Project Overview

"Business news in 2026 is still delivered like it's 2005 — static text articles, one-size-fits-all homepage, same format for everyone. We built something that makes people say: I can't go back to reading news the old way."

What We Built

Team Brain Stack has developed an **AI-Native News Experience** — a multi-agent system that fundamentally reimagines how business news is consumed. Rather than displaying the same static content to every reader, our system uses five specialized AI agents that work together to personalize, synthesize, translate, track, and transform news content in real time.

The Five Agents at a Glance

Agent	Name	What it does
Agent 1	Personalization Agent	Ranks and filters news based on user role, interests, and reading context
Agent 2	Briefing Agent	Synthesizes multiple articles on one topic into a single intelligent briefing
Agent 3	Story Arc Tracker	Builds timelines, tracks sentiment shifts, and predicts what happens next
Agent 4	Vernacular Agent	Culturally adapts news into Hindi, Tamil, Telugu, Bengali — not literal translation
Agent 5	Video Script Agent	Transforms any article into a broadcast-quality 60–120 second video script

Problem Statement

Why Today's News Delivery is Broken

Despite massive advances in AI and personalization technology, most business news platforms in 2026 still operate on a fundamentally outdated model. The same homepage is served to every reader regardless of who they are, what they do, or what language they think in. This creates a massive gap between the information available and the information that actually reaches and benefits readers.

1. No Personalization

A mutual fund investor and a rural farmer see the exact same news feed. There is no mechanism to rank, filter, or adapt content based on the reader's profession, interests, or context.

2. Information Fragmentation

A major event like the Union Budget generates 15-20 separate articles. Readers must spend 30+ minutes reading all of them to form a complete picture. There is no system to synthesize this into one coherent briefing.

3. English-Only Barrier

Over 90% of India thinks and communicates in regional languages, yet business news is almost exclusively in English. Existing translation tools provide literal translations that fail to convey financial concepts meaningfully to regional audiences.

4. No Narrative Intelligence

Ongoing stories like corporate controversies, policy changes, or market movements span months or years. No platform helps readers track the arc of these stories — how sentiment has shifted, who the key players are, and what to expect next.

5. Static Format

News is delivered only as text articles. There is no automated mechanism to transform breaking news into video scripts, briefings, or other formats that suit different consumption patterns.

Core Insight: The problem is not a lack of content — India has abundant quality journalism. The problem is that content delivery has not evolved. Our system builds the intelligence layer that sits on top of existing content and makes it work for every individual reader.

System Architecture & Agent Design

Multi-Agent Architecture Overview

Our system follows a multi-agent architecture where each agent is an independent, specialized AI module with its own system prompt, input schema, and output format. Agents can be called independently or orchestrated together for complex tasks. All agents are powered by LLaMA 3.3 70B via the Groq API.

Layer	Component	Role
Input	User Interface (HTML/JS)	Accepts user inputs: profile, articles, topic, language preference
Orchestration	FastAPI Backend	Routes requests to appropriate agents, runs agents in parallel where possible
Intelligence	5 Specialized AI Agents	Each agent has a unique system prompt defining its role and output schema
AI Engine	Groq API + LLaMA 3.3 70B	Processes all agent requests, returns structured JSON responses
Output	Rendered UI Cards	Parsed JSON rendered as interactive news cards, timelines, scripts

Agent Design Details

Agent 1 — Personalization Agent

- Input: User profile (name, role, interests) + list of articles
- Process: Scores each article against user profile using semantic matching
- Output: Ranked articles with relevance score (0-100) and personalized explanation
- Key innovation: Role-aware ranking — a CFO and a student see completely different feeds

Agent 2 — News Navigator / Briefing Agent

- Input: Topic name + multiple articles (up to 10)
- Process: Synthesizes all articles, extracts key sections, identifies players and sentiment
- Output: Single briefing with headline, TL;DR, insights, contrarian view, follow-up questions
- Key innovation: Replaces 8 articles with 1 explorable briefing — saves 25+ minutes

Agent 3 — Story Arc Tracker

- Input: Story name + chronological list of events/articles
- Process: Builds narrative arc, tracks sentiment shift, identifies story phase
- Output: Timeline, sentiment arc, contrarian perspectives, predictions
- Key innovation: Turns months of coverage into a single navigable story map

Agent 4 — Vernacular Agent

- Input: Article title + content + target language + reader context
- Process: Cultural adaptation using local analogies, idioms, and context-specific impact
- Output: Translated title, adapted article, local analogy, key terms explained
- Key innovation: Adapts financial concepts using local metaphors (farming, local markets)

Agent 5 — Video Script Agent

- Input: Article title + content + target duration (60-120 seconds)
- Process: Structures content into broadcast segments with visual directions
- Output: Hook, timestamped segments, narration, visual directions, closing line, stats
- Key innovation: Any article becomes a broadcast-ready script in under 10 seconds

Technology Stack

Core Technologies

Category	Technology	Purpose
AI Engine	Groq API + LLaMA 3.3 70B	Powers all 5 agents — fast inference, free tier available
Backend	Python + FastAPI	REST API layer, agent orchestration, parallel execution with asyncio
Frontend	HTML5 + CSS3 + Vanilla JS	Single-file demo UI with 5 tabbed agent interfaces
Data Format	JSON (structured outputs)	All agent responses are structured JSON — easy to parse and render
Fonts	Google Fonts (Playfair Display, IBM Plex Mono)	Editorial aesthetic matching ET brand identity
Deployment	Browser-based (no server needed for demo)	Single HTML file runs entirely in the browser for hackathon demo

Why Groq + LLaMA 3.3 70B?

- Free tier with generous rate limits — suitable for hackathon and prototyping
- LLaMA 3.3 70B produces high-quality structured JSON outputs reliably
- Extremely low latency (typically under 2 seconds per agent call)
- Open-weight model — can be self-hosted for production deployment
- Supports all Indian languages required for the Vernacular Agent

System Requirements

Requirement	Specification
Python Version	Python 3.10+
Key Dependencies	fastapi, uvicorn, groq, pydantic, httpx
API Key	Groq API Key (free at console.groq.com)
Browser	Any modern browser (Chrome, Firefox, Edge)

Internet	Required for API calls to Groq
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How We Are Different

Traditional News Platform vs. Our System

Feature	Traditional Platform	Our AI System
News Feed	Same for all users	Personalized by role + interests
Multiple articles on 1 topic	Read all 8 separately	1 synthesized briefing with insights
Language	English only	4 regional languages, culturally adapted
Story tracking	Manual — read 50 articles	Auto timeline + sentiment arc + predictions
Content formats	Text articles only	Articles + Briefings + Video scripts
User intelligence	No understanding of reader	Role-aware, context-aware delivery
Regional coverage	Same national content	Local impact explained per reader type
Data usage	Stores data only	Uses data to generate intelligence

Key Differentiators

Cultural Intelligence, Not Just Translation

Our Vernacular Agent does not use dictionary-based translation. It uses LLaMA 3.3 70B to understand the financial concept, then re-explains it using analogies familiar to the target reader — a farmer understands repo rate through crop loan analogies, not banking terminology.

Agent Specialization for Quality

Each of our 5 agents has a unique, carefully crafted system prompt that makes it an expert in its specific task. This specialization produces significantly higher quality outputs than a single general-purpose AI prompt.

Real-Time Synthesis at Scale

Our Briefing Agent can synthesize up to 10 articles into a single coherent briefing in under 5 seconds. This is not summarization — it extracts cross-article insights, identifies contrarian views, and generates intelligent follow-up questions.

Narrative Intelligence

The Story Arc Tracker is unique in the news industry. It understands stories as living narratives with phases (emerging → developing → peak → resolving → resolved), tracks how sentiment has shifted over time, and surfaces perspectives that mainstream coverage misses.

Future Scope & Roadmap

The current implementation is a proof-of-concept demonstrating the core multi-agent architecture. The following enhancements are planned for production deployment:

Phase	Feature	Description
Phase 2	Live ET Article Integration	Connect directly to ET RSS feeds and APIs for real-time news processing
Phase 2	User Profile Learning	Track reading history to improve personalization over time
Phase 3	Recommendation Engine	ML-based book/article recommendation using user borrowing and review data
Phase 3	Audio Briefings	Convert text briefings to audio using TTS for commute listening
Phase 4	Actual Video Generation	Use AI video tools to generate actual news videos from scripts
Phase 4	More Regional Languages	Expand to Marathi, Gujarati, Kannada, Malayalam, Punjabi

Vision: Every person in India — regardless of their language, profession, or level of financial literacy — deserves access to business news that is relevant, understandable, and actionable for their specific life. Our multi-agent system is the foundation for that vision.

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Thank you for the opportunity to reimagine how India reads business news.